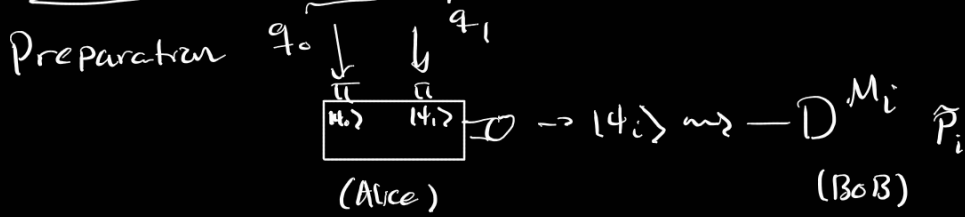


Mixed States: $\rightarrow$ prior probabilities



Dynamics: $U^\dagger U = U U^\dagger = I$
Measurement: $P_i = \langle \psi | M_i | \psi \rangle$
Observable: $A = A^\dagger$

$|\psi\rangle \longrightarrow P_i = \langle \psi | M_i | \psi \rangle = \text{tr}(M_i |\psi\rangle\langle\psi|)$

$\hat{P}_i = q_0 \langle \psi_0 | M_i | \psi_0 \rangle + q_1 \langle \psi_1 | M_i | \psi_1 \rangle \quad | \quad \text{Tr}(CA) = \sum_i \lambda_i = \sum_i \langle i | A | i \rangle$

$\therefore \langle \psi | A | \psi \rangle = \text{Tr}(A |\psi\rangle\langle\psi|) = \sum_i \langle i | A | \psi \rangle \langle \psi | i \rangle$
 $= \langle \psi | \left(\sum_i |i\rangle\langle i| \right) A | \psi \rangle$
 $= \langle \psi | A | \psi \rangle$

$\langle \psi | A | \psi \rangle = \text{Tr}(A |\psi\rangle\langle\psi|)$

$\therefore \hat{P}_i = q_0 \text{Tr}(M_i |\psi_0\rangle\langle\psi_0|) + q_1 \text{tr}(M_i |\psi_1\rangle\langle\psi_1|)$

$= \text{Tr} \left[\underbrace{M_i}_{\text{Measurement}} \underbrace{(q_0 |\psi_0\rangle\langle\psi_0| + q_1 |\psi_1\rangle\langle\psi_1|)}_{\text{State (preparation)}} \right]$

* Mixed State

$\rho = \sum_{i=0}^n q_i |\psi_i\rangle\langle\psi_i|$

Pure state

mixed state

$\rho: \text{pure} \iff \text{tr}(\rho^2) = 1$

Proof: $\text{tr}(\rho) = \sum_{l=1}^n \langle l | \left(\sum_i q_i |\psi_i\rangle\langle\psi_i| \right) | l \rangle$
 $= \sum_i q_i \langle \psi_i | \left(\sum_l |l\rangle\langle l| \right) | \psi_i \rangle$
 $= \sum_i q_i = 1 \quad (\text{true for all types})$

$$\begin{aligned}
\text{Now: } \text{tr}(\rho^2) &= \sum_{\ell} \langle \ell | \left(\sum_i q_i |\psi_i\rangle \langle \psi_i| \right) \left(\sum_j |\psi_j\rangle \langle \psi_j| \right) | \ell \rangle \\
&= \sum_{i,j} q_i q_j \langle i | i \rangle \langle j | \left(\sum_{\ell} |\ell\rangle \langle \ell| \right) | i \rangle \\
&= \sum_{i,j} q_i q_j |\langle i | j \rangle|^2 \\
&= \sum_i q_i^2 \neq 1
\end{aligned}$$

Bloch sphere